

INTERNSHIP REVIEW

YOLOv8s Fine-Tuning on SARDet-100K

YOLOv8s — Architecture Overview

Backbone

Cross Stage Partial Darknet: A feature extractor using **C2f blocks** and **SPPF** (Spatial Pyramid Pooling - Fast) for multi-scale pooling. Outputs feature maps at 3 distinct resolutions.

Neck

Feature Pyramid Network + Path Aggregation Network: A bidirectional aggregator that fuses semantic (top-down) and spatial (bottom-up) contexts, generating processed maps at 64, 32, and 16-pixel scales.

Head

Decoupled Anchor-Free: Separates box regression and classification branches. Eliminates anchors and uses **Non-Maximum Suppression** to resolve duplicate detections.

Input: 512×512 · 11.1M params · One forward pass → boxes + classes

SARDet-100K — Dataset Overview

What is SAR?

- Radar-based imaging — works day/night, through clouds & rain
- Images are greyscale with speckle noise
- Looks very different from natural photos (domain gap)

Dataset Facts:

- 116K Total Images
- 245K Instances
- 10 Source Sensors, 6 Object Classes

Class Distribution (Training Instances)

Class	Train Instances	Challenge
Ship	93,373	Dominant — heavily overrepresented
Aircraft	40,705	Strong scattering signature
Bridge	27,615	Elongated — hard to box
Tank	24,187	Very small (avg 461 px)
Car	9,561	Tiny objects
Harbor	3,306	Rarest — only 2% of instances

How We Fine-Tuned YOLOv8s

1. Convert Labels

- COCO JSON → YOLO .txt format
- Absolute pixel box → normalized cx cy w h

2. Balance Classes

- Cap dominant classes at 15K images each
- Dataset reduced: 94K → 48K images

3. SAR Augmentation

- Rotation $\pm 180^\circ$ (SAR has no up/down)
- No colour jitter (pure greyscale)

4. Train Config

- AdamW, lr=0.001, 2× T4 GPUs
- Mosaic augmentation (disabled last 10 ep)

5. Evaluation

- Val set completely held out
- Best checkpoint saved by mAP@50

Results

Overall Metrics: Before vs After Class-Balanced Undersampling

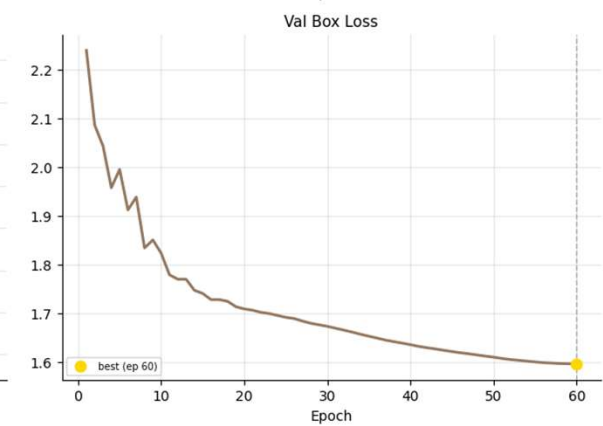
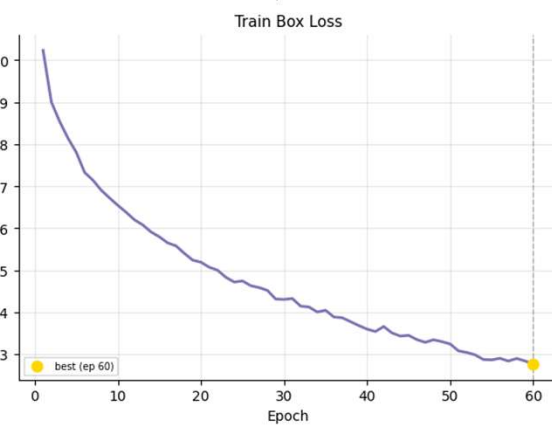
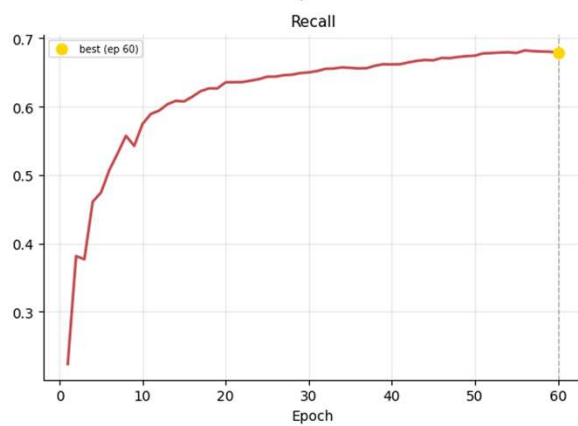
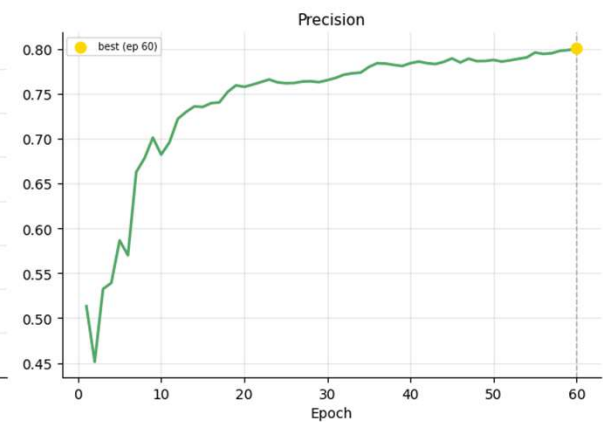
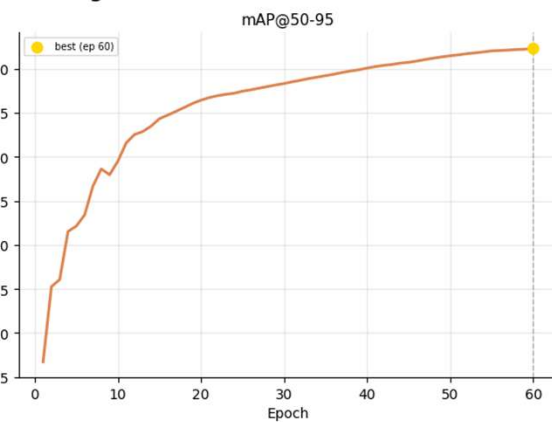
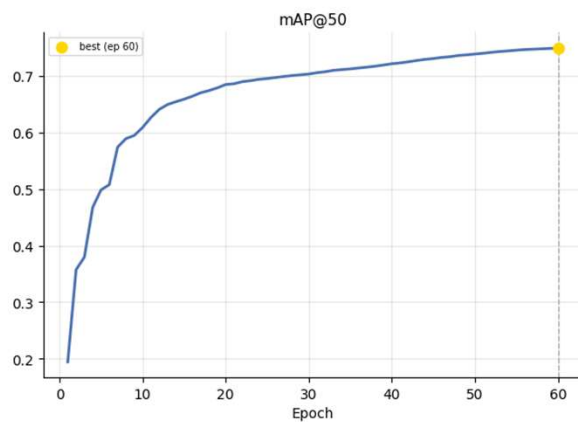
	mAP@50	mAP@50-95	Precision	Recall	Epochs	Train Time
Before	73.45%	41.28%	80.55%	65.42%	50	9.8 h
After	74.91%	42.30%	80.23%	67.98%	60	6.7 h
Change	+1.46	+1.02	-0.32	+2.56	+10	-3.1 h

Per-Class Average Precision (Held-Out Val Set)

Class	Images	Instances	Precision	Recall	AP@50	AP@50-95
Ship	6,441	10,528	85.2%	80.4%	86.6%	45.6%
Aircraft	1,972	5,193	88.1%	90.5%	93.4%	57.9%
Car	104	1,222	82.4%	88.4%	92.8%	55.1%
Tank	156	2,030	78.3%	42.2%	54.0%	31.2%
Bridge	1,747	3,318	71.2%	35.5%	46.3%	18.7%
Harbor	165	404	75.3%	70.8%	76.5%	45.3%
All	10,492	22,695	80.2%	68.0%	74.9%	42.3%

Important Curves

Training Curves -- YOLOv8s v3 on SARDet-100K



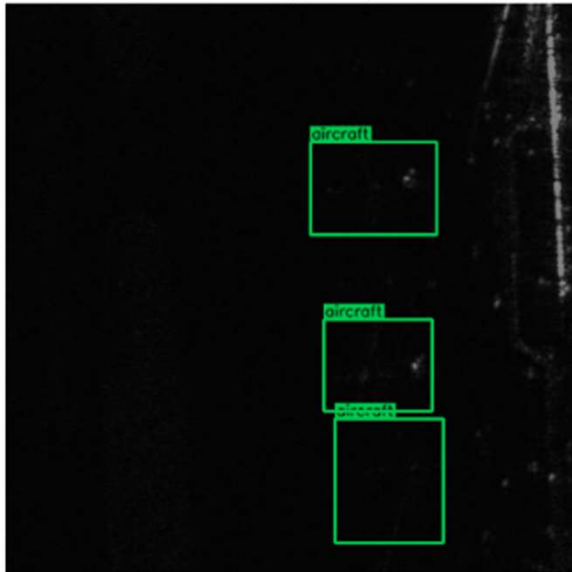
Comparison with State-of-the-Art on SARDet-100K

Method	Backbone	Pretrain	mAP	AP@50	Source
Faster R-CNN	R-50	ImageNet	39.22	70.04	Ni et al., 2025
RetinaNet	R-50	ImageNet	46.48	77.74	Ni et al., 2025
FCOS	R-50	ImageNet	52.52	85.82	Ni et al., 2025
Cascade R-CNN	R-50	ImageNet	53.55	87.33	Ni et al., 2025
GFL	R-50	ImageNet	55.01	85.16	Ni et al., 2025
MSFA (GFL R-50)*	R-50	IN→DOTA→SAR	53.7	84.2	Li et al., 2024
DenoDet V1	R-50	ImageNet	55.88	85.81	Ni et al., 2025
DenoDet V2 (SOTA)	R-50	ImageNet	56.71	85.86	Ni et al., 2025
Ours (YOLOv8s)	CSPDark	COCO	42.3	74.9	This work

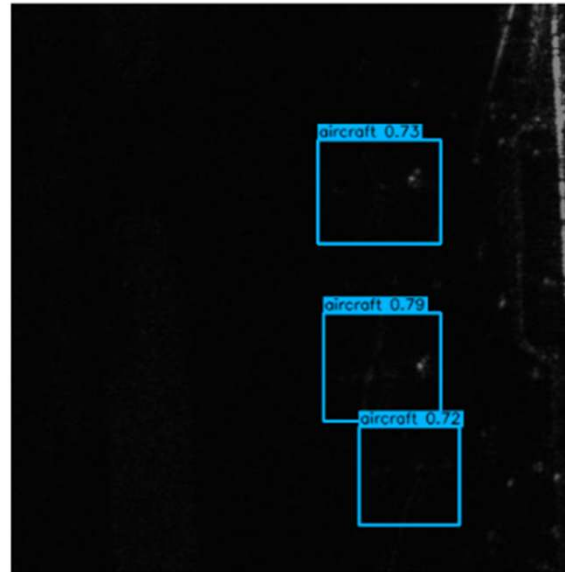
Key Insight: Our YOLOv8s reaches 74.9% AP@50 using only COCO pretraining and simple balanced training — no SAR-domain pretraining or custom modules. SOTA (DenoDet V2) achieves 85.86% with specialised architecture.

Sample outputs

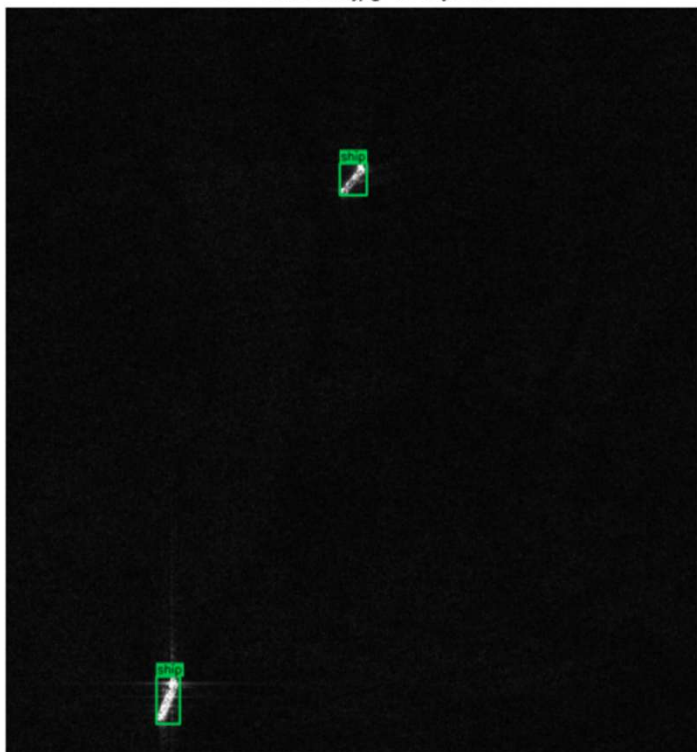
GT -- 0007799.jpg (3 objects)



Predicted -- 3 detections



GT -- 0008426.jpg (2 objects)



Predicted -- 2 detections



Summary

- Base Model: YOLOv8s (COCO pretrained) — no SAR-specific architecture
- Dataset: SARDet-100K · 6 classes · 116K images · severe class imbalance
- Key Trick: Class-capped undersampling at 15K/class → 48K images, +10 epochs
- Augmentation: Rotation $\pm 180^\circ$, flip, brightness — SAR-appropriate, no colour jitter
- Hardware: 2× Tesla T4, 60 epochs, 6.7 hours
- Result: mAP@50 = 74.9% · Aircraft: 93.4% · Bridge: 46.3%